Rahil Parikh

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Robotics engineer specializing in real-time control software for safety-critical hardware, now extending that foundation into learned control. Three years shipping production C++ control systems on embedded Linux — a 47,000-line closed-loop stack running daily at customer research facilities, with FPGA safety interlocks and sub-75 ms deterministic guarantees. Prior work spans whole-body balance control on a bipedal humanoid and deep RL locomotion policies in NVIDIA Isaac Lab. B.S. Robotics Engineering & Computer Science, WPI (3.8); incoming M.S.E. Robotics, Penn GRASP, Fall 2026.

Experience

Control Systems Software Engineer — Sep 2023 – Apr 2026
Bridge12 Technologies, Natick, MA

- Author and owner of a 47,000-line Qt/C++ real-time control system coordinating 8+ heterogeneous instruments through a multi-device finite state machine; deployed in production and operated daily by non-engineers at customer research facilities
- Enforced machine safety through FPGA hardware interlocks arbitrating against the software control law — hard constraints override normal operation deterministically within the loop's timing budget, protecting high-power instrumentation from damage
- Cut closed-loop latency 70% (5.0 s → 1.5 s) by replacing synchronous polling with a ZeroMQ asynchronous publisher and Kalman-filtered distributed state estimation, raising usable control bandwidth
- Delivered hard real-time search-and-track over 100,000+ data points in under 75 ms via spatial hashing and k-d trees, holding deterministic timing under full system load
- Engineered Yocto-based Linux distributions for NVIDIA Jetson Nano and Orin NX, tuning kernel and boot configuration to eliminate latency jitter on embedded control hardware
- Led on-site deployment, debugging, and operator training across 5+ research sites (incl. Columbia University, Havemeyer Hall), converting ambiguous physics requirements into shippable closed-loop systems

Software Engineering Intern — May – Oct 2022
Ekotrope, Boston, MA

- Raised MariaDB transaction throughput 20% through query and schema redesign on a physics-based building energy modeling backend; reduced modeling defect rate 18% with JUnit suites validated against analytical baselines

Education

University of Pennsylvania, Philadelphia, PA — Incoming Fall 2026
M.S.E. in Robotics (GRASP Lab), School of Engineering and Applied Science — Expected 2028

Worcester Polytechnic Institute, Worcester, MA — Aug 2019 – May 2023
B.S. in Robotics Engineering and Computer Science, GPA: 3.8/4.0

- **Major Qualifying Project (senior capstone)**: bipedal humanoid with reactive balancing and partial walking — see Projects
- **Coursework**: Robot Dynamics, Feedback Control, Linear Systems, Motion Planning, Embedded Systems, Computer Vision, Machine Learning, Algorithm Design
- WPI Presidential Scholarship; Dean's List; SWEET Fellowship (research dedication and technical communication); Upsilon Pi Epsilon

Research & Projects

Bipedal Humanoid Robot — *Major Qualifying Project | C++, ROS, Gazebo* — 2022 – 2024

- Led a 12-person interdisciplinary team building a bipedal humanoid for dynamic walking; secured $15,000 in prototype funding through a faculty advisory pitch cycle
- Implemented inverse-kinematics gait generation with ZMP-based stability feedback, achieving sub-50 ms reactive balance response on hardware and sustained partial walking
- Owned the hardware–software interface end to end: actuator calibration, sensor fusion, and the real-time loop coupling balance feedback to joint commands

Quadruped Locomotion via Deep RL | *Isaac Lab, RSL-RL, PPO* — Nov 2025 – Present

- Trained four PPO locomotion policies on AnymalC across a flat/slope/stairs terrain curriculum, achieving verified trotting gaits in all four (duty cycle 0.33–0.62, diagonal foot correlation −0.56 to −0.90) under held-out-terrain gait analysis

- Designed a leave-one-out reward ablation quantifying the reward-vs-behavior gap: `terrain_clearance` proved most valuable (−20.2% task return when removed), while removing `foot_slip` raised reward +32.6% yet halved walking distance
- Diagnosed a silent checkpoint-loading failure in rsl-rl v5 where mismatched state-dict keys masked near-zero locomotion behind a standing bonus; resolved across the pipeline via a shared runner utility

Active Perception for Legged Locomotion | *Isaac Lab, PyTorch* Apr 2026 – Present

- Built a policy-controlled pan-tilt depth camera on a quadruped with an information-gain reward grounded in a terrain-predictor's error on future foothold heights, evaluated across five ablation conditions

Autonomous Mobile Robot Navigation | *C++, ROS, LiDAR, AMCL* 2021

- Built a full autonomy stack (SLAM, particle-filter localization, A*/`move_base` planning) with Kalman-filtered LiDAR–odometry fusion; demonstrated collision-free indoor navigation as lead software developer

Robotic Arm Vision & Motion Planning | *C++, MATLAB, OpenCV* 2021

- Built a forward/inverse-kinematics pipeline with an OpenCV pose-estimation front end driving a 6-DOF manipulator to autonomous color-sorted pick-and-place

Technical Skills

Control & Estimation: real-time and safety-critical control, deterministic loop design, feedback control, PID, Kalman filtering, state estimation, contact dynamics, ZMP balance control, inverse kinematics, trajectory optimization, MPC formulation, SO(3)/SE(3) Lie groups

Embedded & Systems: C++ (production, multithreaded), NVIDIA Jetson (Nano, Orin NX), embedded Linux, Yocto, FPGA hardware interlocks, Qt/QML, ZeroMQ, ROS/ROS2, Docker, Git

Robot Learning & Simulation: NVIDIA Isaac Lab, Isaac Gym, MuJoCo, Gazebo, RSL-RL, PPO, reward shaping, terrain curricula, sim-to-real transfer, domain randomization

Perception: PyTorch, OpenCV, point cloud processing, LiDAR sensor fusion, camera calibration, pose estimation

Languages: C++, Python, MATLAB, CUDA, Java, SQL